

Lecture 9

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1 Reed Muller Codes

Recall the recursive construction of a Reed Muller $RM(r, m)$ as follows:

$$RM(r, m) = \{(u, u + v), u \in RM(r, m - 1), v \in RM(r - 1, m - 1)\}$$

It follows that the generator matrix $G_{r,m}$ will be of the form:

$$\begin{bmatrix} G_{r,m-1} & G_{r,m-1} \\ 0 & G_{r-1,m-1} \end{bmatrix}$$

We will now prove another property of Reed Muller codes:

Theorem 1 *Dual Code of $RM(r, m)$ is $RM(m - r - 1, m)$*

Proof Consider $RM(0, m)$. Recall that this is also the Repetition code and

$$\dim(RM(0, m)) = 1$$

We also know that $RM(m - 1, m)$ will be the simple parity check code with

$$K = \dim(RM(m - 1, m)) = \sum_{i=0}^{m-1} \binom{m}{i} = 2^m - 1$$

We have shown previously that the repetition code generator matrix is orthogonal to the simple parity check code generator matrix. We also have

$$\dim(RM(0, m)) + \dim(RM(m - 1, m)) = 1 + 2^m - 1 = n$$

We will now prove the theorem using induction:

Base Case: $RM(0, 2)$ - As discussed above, this is the repetition code and the generator matrix will be:

$$G = \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}$$

$RM(1, 2)$ - As discussed above, this is the parity check matrix and its generator matrix will be:

$$\hat{G} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

It is clear that $G \perp \hat{G}$ and $\dim(G) + \dim(\hat{G}) = 4$. Assume that the statement is true for $m - 1$. The consequence of this is:

- $G_{r,m-1} \perp G_{m-r-2,m-1}$
- $G_{r-1,m-1} \perp G_{m-r-1,m-1}$

We have:

$$G_{r,m} = \left[\begin{array}{c|c} a & a \\ \hline G_{r,m-1} & G_{r,m-1} \\ 0 & G_{r-1,m-1} \\ \hline 0 & c \end{array} \right] \quad G_{m-r-1,m} = \left[\begin{array}{c|c} b & b \\ \hline G_{m-r-1,m-1} & G_{m-r-1,m-1} \\ 0 & G_{m-r-2,m-1} \\ \hline 0 & d \end{array} \right]$$

We first show that:

$$\begin{aligned} & \dim(G_{r,m}) + \dim(G_{m-r-1,m}) = n = 2^m \\ \Leftrightarrow & \sum_{i=0}^r \binom{m-1}{i} + \sum_{i=0}^{r-1} \binom{m-1}{i} + \sum_{i=0}^{m-r-1} \binom{m-1}{i} + \sum_{i=0}^{m-r-2} \binom{m-1}{i} = 2^m \\ \Leftrightarrow & \sum_{i=0}^r \binom{m-1}{i} + \sum_{i=0}^{r-1} \binom{m-1}{i} + \sum_{i=r}^{m-1} \binom{m-1}{i} + \sum_{i=r+1}^{m-1} \binom{m-1}{i} = 2^m \\ \Leftrightarrow & \sum_{i=0}^{m-1} \binom{m-1}{i} + \sum_{i=0}^{m-1} \binom{m-1}{i} = 2^m \\ \Leftrightarrow & 2^{m-1} + 2^{m-1} = 2^m \\ \Leftrightarrow & 2^m = 2^m \end{aligned}$$

We will now show orthogonality:

$$\dim(G_{r,m}) \perp \dim(G_{m-r-1,m})$$

We have the following cases:

1. $a|a \perp b|b$: Note that $a.b + a.b = 0$ for $GF(2)$
2. $a|a \perp 0|d$: Follows from orthogonality of $m-1$, since we have $G_{r,m-1} \perp G_{m-r-2,m-1}$
3. $0|c \perp b|b$: Follows from orthogonality of $m-1$, since we have $G_{r-1,m-1} \perp G_{m-r-1,m-1}$
4. $0|c \perp 0|d$: For this, note that $G_{r-1,m-1}$ is a subset of $G_{r,m-1}$. Hence $G_{r-1,m-1} \perp G_{m-r-2,m-1}$

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1.1 Another construction for Reed Muller codes

Consider:

$$G^* = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

We define the Kronecker product of G^* as follows:

$$G^* \otimes G^* = \begin{bmatrix} G_{11}^* \cdot G^* & G_{12}^* \cdot G^* \\ G_{21}^* \cdot G^* & G_{22}^* \cdot G^* \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Similarly, we can define the m - fold Kronecker product of G^* can be defined which will be of dimension $2^m \times 2^m$. Note that, this matrix will contain $\binom{m}{l}$ rows of weight 2^{m-l} . It can be shown that $RM(r, m)$ is constructed by taking rows of weight $\geq 2^{m-l}$

1.2 Decoding of RM codes

There exists an efficient decoding algorithm for $RM(r, m)$ which can correct upto $2^{m-r}/2 - 1$ errors (i.e. $2^{m-r-1} - 1$ errors), the idea being using majority decoding iteratively.

Assume that we have such a decoder for the $(r-1)^{th}$ order which can correct upto $2^{m-(r-1)}/2 - 1$ errors. Now we construct a decoder for the r^{th} order:

$r = 0$: Repetition code $\rightarrow$ Majority decoder can correct upto $2^m/2 - 1$ errors.

$$k = \sum_{i=0}^r \binom{m}{i}$$

$$G = \begin{bmatrix} G_1 \\ G_2 \\ G_3 \\ \vdots \\ G_r \end{bmatrix}$$

Let $u_0 u_1 u_2 \dots u_{k-1}$ be the message symbols, written as $\bar{u}_0 \bar{u}_1 \dots \bar{u}_r$ where $\bar{u}_i$ corresponds to the block of message symbols being multiplied with G_i consisting of $\binom{m}{i}$ symbols.

$$c = \begin{bmatrix} \bar{u}_0 & \bar{u}_1 & \dots & \bar{u}_r \end{bmatrix} \begin{bmatrix} G_1 \\ G_2 \\ \vdots \\ G_r \end{bmatrix}$$

$$v = c + e$$

(e being the error term)

Now if $\bar{u}_r$ is estimated correctly,

$$v' = c - \bar{u}_r \cdot G_r + e = \begin{bmatrix} \bar{u}_0 & \bar{u}_1 & \dots & \bar{u}_{r-1} \end{bmatrix} \begin{bmatrix} G_0 \\ G_1 \\ \vdots \\ G_{r-1} \end{bmatrix} + e$$

Now this has order $r - 1$, (essentially a $\text{RM}(r - 1, m)$ code) and can be decoded using Majority decoding as per the induction assumption. Hence goal: estimate $\bar{u}_r$ correctly!

$$\bar{u}_r = \begin{bmatrix} \cdot & \cdot & \cdot & u_{k-1} \end{bmatrix}$$

u_{k-1} multiplied by last row of G_r , let the received word be v :

$$v = \begin{bmatrix} v_0 & v_1 & \dots & v_{2^m-1} \end{bmatrix}$$

Decoding rule:

Create 2^{m-r} parity checks each consisting of 2^r received symbols with no overlap.

→ In each parity, u_{k-1} should contribute only once to symbols in parity.

→ All other message symbols contribute an even no. of times.

→ Idea: For no noise, $p_i = u_{k-1}$. 2^{m-r} parity checks using majority decoder can recover upto $(2^{m-r}/2 - 1)$ errors. (p_i has been defined in the following example:)

Example: (2,4) RM code

This is essentially a (16, 11) code (check!)

$$\begin{bmatrix} u_0 & u_1 & \dots & u_{10} \end{bmatrix} \longrightarrow \begin{bmatrix} \bar{u}_0 & \bar{u}_1 & \bar{u}_2 \end{bmatrix}$$

$$\begin{bmatrix} \bar{u}_0 & \bar{u}_1 & \bar{u}_2 \end{bmatrix} \begin{bmatrix} G_0 \\ G_1 \\ G_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & \dots \\ 0 & 0 & 0 & 0 & \dots \\ 0 & 0 & 0 & 0 & \dots \\ 0 & 0 & 1 & 1 & \dots \\ 0 & 1 & 0 & 1 & \dots \\ 0 & 0 & 0 & 0 & \dots \\ 0 & 0 & 0 & 0 & \dots \\ 0 & 0 & 0 & 0 & \dots \\ 0 & 0 & 0 & 0 & \dots \\ 0 & 0 & 0 & 0 & \dots \\ 0 & 0 & 0 & 1 & \dots \end{bmatrix}$$

Create parity checks p_i as follows:

(Notice that p_1 indeed satisfies the conditions of a parity check for u_{10})

$$p_1 = v_0 + v_1 + v_2 + v_3$$

$$p_2 = v_4 + v_5 + v_6 + v_7$$

$$p_3 = v_8 + v_9 + v_{10} + v_{11}$$

$$p_4 = v_{12} + v_{13} + v_{14} + v_{15}$$

We want to construct 2^{m-r} parity checks each consisting of 2^r bits.
(Now try the same for u_9 !)

Repetition code: $(n, 1, n)$ code

Simple Parity check code: $(n, n-1, 2)$ code

Hamming code: $(2^m - 1, 2^m - m - 1, 3)$ code ($m \geq 1$)

Reed Muller code: $(2^m, \sum_{i=0}^r \binom{m}{i}, 2^{m-r})$ code ($m \geq 1, 0 \leq r \leq m$)

$\implies$ There are restrictions on (n, k, d) .

One such constraint is the **Hamming Bound/ Sphere Packing Bound**:

For any (n, M, d) code over $GF(q)$,

$$M \leq \frac{q^n}{vol_q(n, \lfloor \frac{d-1}{2} \rfloor)}$$

For a linear code: $M = q^k$.